

A finite criterion for complete information order with a one-dimensional source kernel

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Abstract

For finite-dimensional quantum channels, complete symmetric logarithmic derivative order admits a necessary and sufficient common Hermiticity-preserving observable lift whenever the source adjoint has kernel dimension at most one. The lift is characterized by one finite semidefinite program. A reference of dimension equal to the real dimension of the target Hermitian algebra suffices to test the SLD order. The proof uses an exact null-vector block score to make one prescribed combination of complex lift coefficients Hermitian without a limiting Gram penalty. Convex duality then supplies precisely the combination needed in the one-dimensional-kernel case. The same argument excludes every strict dual obstruction with imaginary-gradient rank at most one. Combined with complete-order universality, the finite program also decides unrestricted complete relative-entropy and interior Petz Rényi order. Numerical conditioning and entropy-witness reference sizes require separate qualifications.

1 Definitions and the finite criterion

Let $\mathcal{P} : M_a \rightarrow M_e$ and $\mathcal{Q} : M_a \rightarrow M_b$ be completely positive trace-preserving maps. Write $p = \mathcal{P}^*$ and $q = \mathcal{Q}^*$. Compress the output spaces to the supports of $\mathcal{P}(I)$ and $\mathcal{Q}(I)$, so that these matrices are strictly positive. All kernel dimensions below are measured after this compression. Write $p_n = p \otimes \text{id}_n$, with analogous notation for other maps.

For a positive definite matrix σ and Hermitian H , define

$$\mathcal{S}_\sigma(X) = \frac{\sigma X + X \sigma}{2}, \quad g_\sigma(H) = \text{Tr } H \mathcal{S}_\sigma^{-1}(H).$$

Complete SLD order $\mathcal{P} \succeq_{c,\text{SLD}} \mathcal{Q}$ means

$$g_{\mathcal{P}_n(\rho)}(\mathcal{P}_n(H)) \geq g_{\mathcal{Q}_n(\rho)}(\mathcal{Q}_n(H)) \quad (1)$$

for every finite reference dimension n , every faithful input state ρ , and every Hermitian tangent H with $\text{Tr } H = 0$. The trace condition can be dropped: writing $H = H_0 + c\rho$ adds the same c^2 to both metrics. Outputs are evaluated on their fixed effective supports.

A common HP observable lift is a complex-linear unital map $J : M_b \rightarrow M_e$ preserving adjoints, such that $pJ = q$ and

$$p_n(J_n(B)^2) \leq q_n(B^2) \quad \text{for every } n \text{ and every Hermitian } B \in M_b \otimes M_n. \quad (2)$$

The map J is not required to be positive.

Fix a Hermitian basis $I, B_1, \dots, B_k$ of M_b , where $k = b^2 - 1$. Define

$$G_Q = [q(B_i B_j)]_{i,j=1}^k.$$

Choose a Stinespring isometry $V : \mathbb{C}^a \rightarrow \mathbb{C}^e \otimes \mathbb{C}^r$ for p , so $p(A) = V^*(A \otimes I_r)V$. For Hermitian variables $C_1, \dots, C_k$ put

$$W(C) = \begin{bmatrix} (C_1 \otimes I_r)V & \cdots & (C_k \otimes I_r)V \end{bmatrix}.$$

The finite feasibility problem is

$$\boxed{C_i = C_i^*, \quad p(C_i) = q(B_i), \quad \begin{bmatrix} G_Q & W(C)^* \\ W(C) & I_{er} \end{bmatrix} \succeq 0.} \quad (3)$$

Its block LMI has size $ak + er$. The Schur complement identifies it with

$$[p(C_i C_j)]_{i,j=1}^k \leq [q(B_i B_j)]_{i,j=1}^k. \quad (4)$$

When the reconstruction constraints are solved first and $\dim_{\mathbb{C}} \ker p = 1$, only k real affine lift parameters remain.

Theorem 1.1. *Assume $\dim_{\mathbb{C}} \ker p \leq 1$. The following are equivalent:*

1. $\mathcal{P} \succeq_{c, \text{SLD}} \mathcal{Q}$.
2. SLD comparison holds at reference dimension b^2 .
3. The finite semidefinite feasibility problem (3) is feasible.
4. A common HP observable lift satisfying (2) exists.

For a classical target with m output letters, replace b^2 by m and use a Hermitian basis of the target diagonal algebra. More generally the reference bound is the real dimension of the Hermitian part of the target finite-dimensional C^* -algebra.

If the target algebra is one dimensional, its only unital lift is $J(cI) = cI$ and the assertion is immediate. Below $k \geq 1$.

2 The score formulation

Lemma 2.1 (Fixed-reference score comparison). *At a fixed reference dimension n , SLD comparison is equivalent to the following statement: every Hermitian target score B has a Hermitian source score A such that*

$$p_n(A) = q_n(B), \quad p_n(A^2) \leq q_n(B^2). \quad (5)$$

Proof. The SLD variational formula is

$$g_{\mathcal{P}_n(\rho)}(\mathcal{P}_n(H)) = \sup_{A=A^*} \{2 \operatorname{Tr} H p_n(A) - \operatorname{Tr} \rho p_n(A^2)\}.$$

Thus (5) implies metric comparison. Conversely, metric comparison first gives $\operatorname{im} q_n \subseteq \operatorname{im} p_n$: otherwise a Hermitian tangent annihilating $\operatorname{im} p_n$ but not $\operatorname{im} q_n$ contradicts the comparison. Fix B and put $x = q_n(B)$, $y = q_n(B^2)$. The affine set $\mathcal{F} = \{A = A^* : p_n(A) = x\}$ is nonempty. Quadratic duality at every faithful ρ gives

$$\begin{aligned} \inf_{A \in \mathcal{F}} \operatorname{Tr} \rho p_n(A^2) &= \sup_H \{2 \operatorname{Tr} H x - g_{\mathcal{P}_n(\rho)}(\mathcal{P}_n(H))\} \\ &\leq \sup_H \{2 \operatorname{Tr} H x - g_{\mathcal{Q}_n(\rho)}(\mathcal{Q}_n(H))\} \leq \operatorname{Tr} \rho y. \end{aligned}$$

The source quadratic is coercive since $\mathcal{P}_n(\rho) > 0$, so the quadratic duality is exact.

Set $f(A, \rho) = \text{Tr } \rho[p_n(A^2) - y]$. The inequality $\inf_A f(A, \rho) \leq 0$ extends from faithful to all density matrices: replace ρ by $(1-t)\rho + t\rho_0$ with fixed faithful ρ_0 , use $f(A, \rho_0) \geq -\text{Tr } \rho_0 y$, and let $t \downarrow 0$. Convex-concave minimax, with the density matrices as the compact variable, yields $\inf_{A \in \mathcal{F}} \max_\rho f(A, \rho) \leq 0$. The maximum includes ρ_0 , hence is coercive in A . Its minimum is attained and supplies (5). $\square$

Lemma 2.2 (Finite Gram criterion). *The finite program (3) is feasible if and only if a common HP observable lift exists.*

Proof. Given feasible C_i , define $J(I) = I$, $J(B_i) = C_i$, and extend complex linearly. For a Hermitian amplified score $B = I \otimes R_0 + \sum_i B_i \otimes R_i$, all constant and linear terms in its square defect cancel by $p(C_i) = q(B_i)$. The remaining defect is

$$\sum_{i,j} (q(B_i B_j) - p(C_i C_j)) \otimes R_i R_j \succeq 0,$$

by compression of the positive coefficient Gram matrix. Conversely, apply (2) to the Hermitian star score with off-diagonal entries B_i . Its lower principal square block is exactly (4). $\square$

3 One Hermitian combination without a Gram penalty

Lemma 3.1 (Null-vector block score). *Suppose SLD comparison holds at reference size $k+1$. For any Hermitian tuple $B_1, \dots, B_k$ and any real unit vector $a \in \mathbb{R}^k$, there are possibly non-Hermitian matrices C_i such that*

$$p(C_i) = q(B_i), \quad [p(C_i C_j^*)] \leq [q(B_i B_j)], \quad \sum_i a_i C_i \text{ is Hermitian.} \quad (6)$$

Proof. Let B denote the column with entries B_i , and set

$$S = \sum_i a_i B_i, \quad F_{ij} = B_i a_j + a_i B_j - a_i a_j S.$$

Then $F = F^*$ and $Fa = B$. For $\varepsilon > 0$ define the Hermitian target score

$$T_\varepsilon = \begin{bmatrix} S/\varepsilon & -B^* \\ -B & \varepsilon F \end{bmatrix}. \quad (7)$$

It annihilates $v \otimes \mathbb{C}^b$, where $v = (\varepsilon, a)$. By Lemma 2.1, choose a feasible Hermitian source score

$$A_\varepsilon = \begin{bmatrix} D_\varepsilon & -C_\varepsilon^* \\ -C_\varepsilon & E_\varepsilon \end{bmatrix}.$$

Since $0 \leq p_{k+1}(A_\varepsilon^2) \leq q_{k+1}(T_\varepsilon^2)$, its quadratic form vanishes on $v \otimes \mathbb{C}^a$. Taking the input trace and using $\mathcal{P}(I) > 0$ shows that A_ε annihilates $v \otimes \mathbb{C}^e$. Therefore

$$\varepsilon D_\varepsilon = C_\varepsilon^* a, \quad E_\varepsilon a = \varepsilon C_\varepsilon. \quad (8)$$

In particular $\sum_i a_i C_{\varepsilon,i}$ is Hermitian, and reconstruction gives $p(C_{\varepsilon,i}) = q(B_i)$.

The lower principal square inequality reads

$$p_k(C_\varepsilon C_\varepsilon^* + E_\varepsilon^2) \leq q_k(BB^* + \varepsilon^2 F^2).$$

Dropping the positive term $p_k(E_\varepsilon^2)$ gives

$$[p(C_{\varepsilon,i} C_{\varepsilon,j}^*)] \leq [q(B_i B_j)] + \varepsilon^2 q_k(F^2). \quad (9)$$

Tracing the diagonal blocks bounds each $C_{\varepsilon,i}$ uniformly for $0 < \varepsilon \leq 1$, because $\mathcal{P}(I) > 0$. A convergent subsequence as $\varepsilon \downarrow 0$ proves (6). $\square$

4 Convex separation and the one dimensional kernel

Proof of Theorem 1.1. Only the implication from fixed-reference SLD order to finite feasibility requires proof. The other implications follow from Lemma 2.2 and the SLD variational formula. Reference comparison at $k+1 = b^2$ implies ordinary comparison, hence $\text{im } q \subseteq \text{im } p$ and Hermitian reconstruction is possible.

For Hermitian reconstructing tuples define

$$D(C) = [p(C_i C_j) - q(B_i B_j)].$$

The function $C \mapsto \lambda_{\max} D(C)$ is convex and coercive. Indeed,

$$\lambda_{\max} D(C) \geq \frac{\text{Tr } D(C)}{ak}, \quad \sum_i \text{Tr } p(C_i^2) = \sum_i \text{Tr } \mathcal{P}(I) C_i^2,$$

which controls the sum of squared Hilbert–Schmidt norms. If the Gram problem were infeasible, its attained minimum would therefore be strictly positive. Minimax supplies a positive matrix weight Ω of trace one with

$$\inf_{C_i = C_i^*, p(C_i) = q(B_i)} \text{Tr } \Omega D(C) > 0. \quad (10)$$

We may assume $\Omega > 0$: mix a strict separator with a sufficiently small multiple of $I/(ak)$. The latter objective has a finite lower bound, so strict separation persists. The resulting objective is coercive and strictly convex in all nonzero lift directions, and its HP minimum is attained.

If $\ker p = 0$, reconstruction is unique. Lemma 3.1 gives the unique tuple with a dominated Gram, a contradiction. Suppose now $\ker p = \mathbb{C}K$, where $K = K^* \neq 0$. Such a Hermitian generator exists because the kernel is closed under adjoints. Let H_i be the Hermitian minimizer in (10). Every complex reconstructing tuple is

$$C_i = H_i + (x_i + iy_i)K, \quad x_i, y_i \in \mathbb{R}.$$

Extend the objective to these tuples by the right Gram,

$$f(C) = \text{Tr } \Omega ([p(C_i C_j^*)] - [q(B_i B_j)]).$$

It is a real convex quadratic. At H its derivative in all real directions x_i vanishes. Let $g \in \mathbb{R}^k$ be its derivative in the imaginary coordinates y_i . Choose $a = g/\|g\|$ if $g \neq 0$, and any real unit a otherwise.

The constraint $\sum_i a_i C_i$ Hermitian is exactly $a \cdot y = 0$. On this affine space the full derivative of f at H vanishes, so convexity implies

$$f(C) \geq f(H) > 0.$$

Lemma 3.1, applied to this chosen a , supplies a tuple in the same affine space with a dominated Gram, hence $f(C) \leq 0$. This contradiction proves feasibility. $\square$

Proposition 4.1 (Finite score witness from a strict dual). *In the preceding proof put $\delta = f(H) > 0$ and $M = \text{Tr } \Omega q_k(F^2) \geq 0$. Every $\varepsilon > 0$ with $\varepsilon^2 M < \delta$ makes (7) a target score with no feasible source completion. If $M = 0$, every $\varepsilon > 0$ works.*

Proof. Any completion obeys the exact Hermitian-combination constraint (8), so its lower-left tuple has $f(C_\varepsilon) \geq \delta$. But (9) gives $f(C_\varepsilon) \leq \varepsilon^2 M$. $\square$

By Lemma 2.1, this finite score obstruction implies a faithful input state and traceless Hermitian tangent violating SLD comparison at the same reference size.

5 A rank one dual extension

For a general Hermitian kernel basis $K_1, \dots, K_r$, let H minimize a strictly separating positive-definite weighted Gram objective. Write

$$C_i = H_i + \sum_{\ell=1}^r (x_{\ell i} + iy_{\ell i})K_{\ell}, \quad G_{\ell i} = \left. \frac{\partial f}{\partial y_{\ell i}} \right|_H.$$

Proposition 5.1. *If $\text{rank}_{\mathbb{R}} G \leq 1$, complete SLD order fails, already at reference size $k+1$.*

Proof. For $G \neq 0$ write $G_{\ell i} = b_{\ell} a_i$ with a a real unit vector. The null-vector gadget imposes $\sum_i a_i y_{\ell i} = 0$ for every ℓ . Thus the full derivative of f at H vanishes on its constrained complex affine space, and the proof above applies unchanged. If $G = 0$, any a works. $\square$

Consequently, a genuine separation of complete SLD order from common HP lifting requires a source-adjoint kernel of dimension at least two. Every positive-definite strict separating weight, evaluated at its attained HP minimizer, must then have imaginary-gradient rank at least two. No reduction of arbitrary separators to rank one is asserted.

6 Relation to entropy orders and precision

The complete-order universality theorem for modular Petz divergences identifies unrestricted complete SLD order with complete Umegaki relative-entropy order and, separately, with complete Petz Rényi order for every fixed $\alpha \in (0, 2)$, including the Umegaki limit at $\alpha = 1$. Here “unrestricted” means comparison on all state pairs and all finite references, without imposing equal reference marginals. It also identifies this preorder with the order of any fixed operator-convex modular Petz generator having finite $f(0+)$ and a nonzero interior representing measure. This is an additional entropy–metric theorem; the proof above concerns the finite observable-lifting step.

Corollary 6.1. *Under the source-kernel hypothesis of Theorem 1.1, the finite program (3) is necessary and sufficient for unrestricted complete Umegaki order, for each fixed interior Petz Rényi order $\alpha \in (0, 2)$, and for each fixed modular Petz generator covered by the preceding universality theorem.*

The equivalence applies first on faithful state pairs and then, by the usual common faithful regularization, with extended support values. It does not cover the Petz Rényi endpoints merely by continuity, and it does not identify numerical contraction coefficients.

The reference bound b^2 is a bound for testing SLD order. It does not by itself give the same bound for direct entropy-pair witnesses: the conversion from a strict SLD violation to an entropy violation can introduce additional reference dimensions depending on the violation margin and spectral conditioning.

Finally, (3) is an exact finite semidefinite characterization. A floating-point solver status or a tiny computed margin is not an exact boundary certificate. Near singular feasibility, validated residual and eigenvalue bounds, exact algebraic certificates, or an appropriate separation promise are needed. No unconditional polynomial bit-complexity or uniform lower bound on the witness margin is claimed.